

Tikendra Pathe

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Professional Summary

AI/ML Engineer with expertise in deep learning, computer vision, and audio processing. Proficient in Python, TensorFlow, and Keras, with a proven track record of developing high-accuracy models (91.6% in medical imaging, 96.2% in voice classification). Skilled in building end-to-end AI solutions using CNNs, transfer learning, and Flask APIs. Experienced in explainable AI (Grad-CAM) and model deployment. Seeking AI/ML internship roles to deliver production-ready solutions.

Skills

Programming: Python, C Programming

ML/AI: Supervised/Unsupervised Learning, Feature Extraction, Model Training, EDA, LLMs, LangChain

Deep Learning: CNNs, DenseNet121, VGG16, EfficientNet, Transfer Learning, Grad-CAM, TensorFlow, Keras

Libraries: NumPy, Pandas, Scikit-learn, Librosa, Matplotlib, Plotly

Web Development SQL, JavaScript, HTML, CSS, React.js, Node.js, Flask, FastAPI, Three.js, Git, GitHub

Education

B.Tech in Computer Science Engineering

Expected 2026

Krishna's Vikash Institute of Technology, Raipur

CSVТУ University

Percentage: 71%

Training

AI/ML Vocational Training

May 2025 – July 2025

IIIT Naya Raipur - Python, Data Science, Machine Learning, Deep Learning

Python Programming Trainee

July 2024 – Aug 2024

CodTech IT Solutions - Automation scripts, debugging, problem solving

Projects

PneumoScan Pro – AI-Based Lung Disease Detection

TensorFlow, Keras, DenseNet121, VGG16, Flask, Grad-CAM, Three.js

- Built AI-powered web platform detecting 5 lung diseases (Pneumonia, COVID-19, Tuberculosis, Lung Opacity, Lung Cancer) from chest X-rays with 91.6% average accuracy in under 5 seconds
- Achieved disease-wise accuracy: Tuberculosis (98%), COVID-19 (94%), Pneumonia (93%), Lung Cancer (84%), Lung Opacity (89%) with average recall of 95.6%
- Implemented Grad-CAM for explainable AI generating heatmaps to visualize disease-affected regions in red
- Developed severity classification (Mild, Moderate, Severe, Critical) and 3D lung visualization using Three.js
- Built Flask web app with telemedicine video consultation, PDF medical reports, patient history tracking, and multi-language support (English, Hindi)

GitHub: github.com/TikendraPathe/PneumoScan-Pro

Voice Gender Classification

Python, Librosa, Scikit-learn, Random Forest, Flask

- Built Random Forest classifier achieving 96.2% accuracy for gender classification from voice recordings (3,168 samples, balanced classes)
- Extracted 40 MFCC features + spectral centroid, zero-crossing rate, and RMS energy using Librosa
- Applied audio preprocessing: silence removal, resampling to 16kHz, and amplitude normalization
- Compared 4 algorithms (Logistic Regression, SVM, Random Forest, XGBoost) with 5-fold cross-validation
- Developed Flask API with web interface for real-time audio upload and instant prediction

GitHub: github.com/TikendraPathe/Voice-Gender-Classification

Certifications

- Python Programming Certification – CodTech IT Solutions
- AI/ML Vocational Training – IIIT Naya Raipur